

Problems Proposed for 2025-26 Pal4Pal

Samuel, Eric, Catherine

October 12, 2025

1 Samuel

1.1 Problem 1 Combo [4 points]

Alice and Bob play a two-player game in which the numbers a and b are initially written on the whiteboard. Starting with Alice, the players take turns subtracting 1 or 2 from one of the current numbers written on the whiteboard. The first player to reach 0 as one of the numbers wins. Given that a and b are chosen randomly between 1 to 101 inclusive, what is the probability that a and b are so that Alice has a winning strategy?

1.2 Problem 2 Number Theory [5 points]

Find the number of primes p for which there exists a positive integer m that is of the form $12k + 2$ such that $48m(47m + 1)$ is p times a perfect square.

2 Eric

2.1 Problem 1 Number Theory [3 points]

Let S be the sum of the squares of the divisors of 480 (including itself). Compute the number of divisors of S .

2.2 Problem 2 Algebra [4 points]

An ellipse in the first quadrant is tangent to both the x -axis and the y -axis. The coordinates of its foci are $(8, 12)$, and $(v, 12)$. Compute v .

3 Catherine

3.1 Problem 1 Geometry [4 points]

Let $\triangle ABC$ be a right triangle with right angle at A and AC the shorter leg. Point D lies inside the triangle such that $\angle ADC = 90^\circ$ and $\angle DCA = \angle ABC$. Extend BD to a point E so that $\angle BEA = 90^\circ$ and $\angle DAE = \angle ABC$. Find the ratio $BD : DE$.

3.2 Problem 2 Algebra [4 points]

Compute $\cos^4 \frac{\pi}{9} + \cos^4 \frac{2\pi}{9} + \cos^4 \frac{3\pi}{9} + \cos^4 \frac{4\pi}{9}$.

3.3 Problem 3 Geometry [5 points]

In triangle ABC , $BC = 6$ and $6 \sin A + 5 \cos B \cos C = 9$. Find the area of ABC .